Amortized Bayesian Experimental Design

Zhanghu Zhao

Supervisor : Samuel Kaski

Probabilistic Machine Learning Group, Aalto University

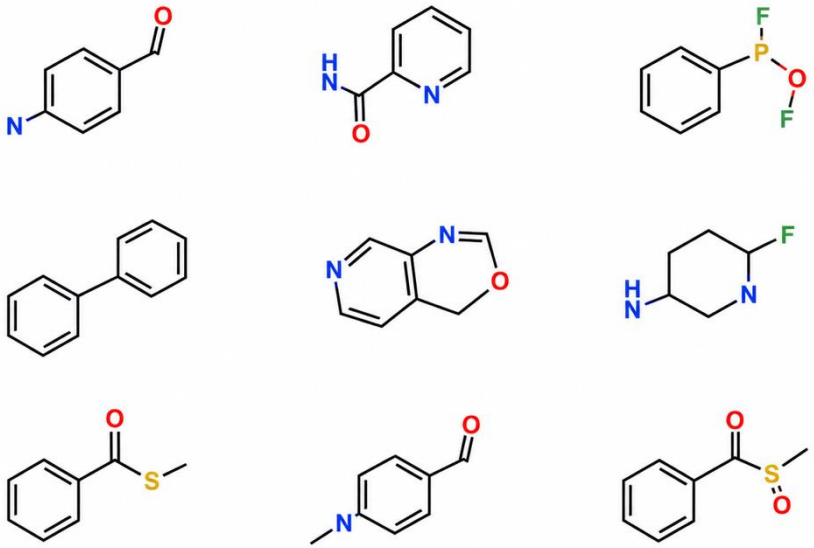
Aalto-yliopisto
Aalto-universitetet
Aalto University

The Critical Need for Data Acquisition

❓How do we select the most informative data to collect under limited resources?

Examples:

- *Personalized Medicine*
- *Materials Discovery*
- *Psychology*
- …



Example: 10,000 candidate molecule designs, with a budget of 10 designs

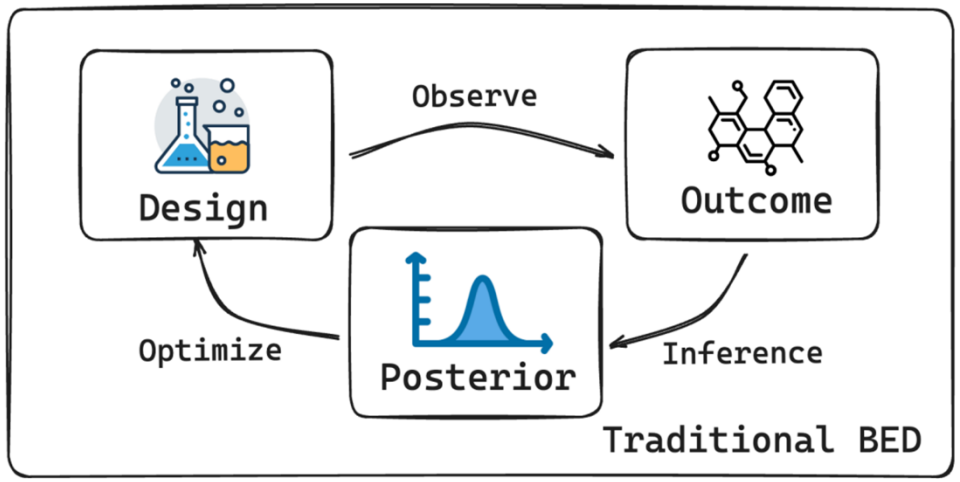
Principled Framework for Data Acquisition

❓How do we select the most informative data to collect under limited resources?

💡**Bayesian Experimental Design (BED):**

- *Optimize experimental design x aimed at maximizing the expected information gain (EIG) about the parameters.*
- *A prior $p(\theta)$, A simulator $p(y|\theta, x)$*
- *Bayes' rule: A posterior $p(\theta|x, y)$*
- $EIG(x) = E_{p(y|x)}[H(\theta) - H(\theta|x, y)]$ **Too slow!**

[1]



[1] Huang, D., Guo, Y., Acerbi, L. and Kaski, S., 2024. Amortized bayesian experimental design for decision-making. *Advances in Neural Information Processing Systems*, *37*, pp.109460-109486.

What is Amortization?

~~Amortization refers to the ability of a conditional distribution approximation to condition on arbitrary instances of the conditioning variable, rather than being specialized to a particular instance.~~ [2]

Too abstract!

[2] Ward, D., Cannon, P., Beaumont, M., Fasiolo, M. and Schmon, S., 2022. Robust neural posterior estimation and statistical model criticism. *Advances in neural information processing systems*, *35*, pp.33845-33859.

What is Amortization?



- *You wash, chop and prepare all ingredients from scratch every time you cook*
- *It takes a lot of time and effort*
- *Even for the same dish, you repeat the same work each time*



- *You spend time upfront to wash and chop ingredients in bulk*
- *For each meal, you just use the prepped ingredients*
- *The upfront effort is reused across many meals, making each meal faster and easier*

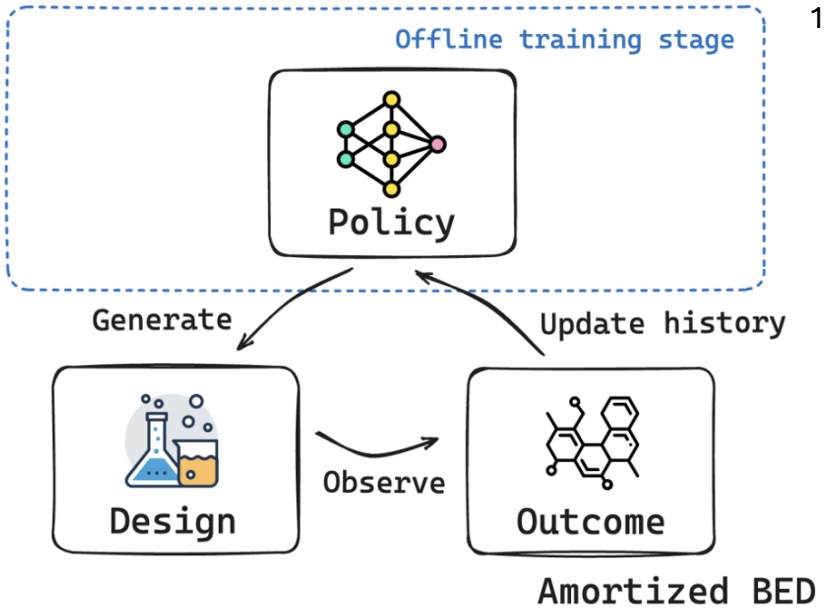
Amortization: Pretrain once, then solve each new task in a single forward pass

Amortization speeds up BED

❓How do we speed up?

💡**Amortization:**

- *Learn a policy π **offline**, a neural network pretrained on a vast number of simulated experiment histories with $\theta \sim p(\theta)$ [1,3,4].*
- *At deployment, π maps a data history directly to the next design, i.e. $x_t = \pi(h_{t-1})$*



1

[1] Huang, D., Guo, Y., Acerbi, L. and Kaski, S., 2024. Amortized bayesian experimental design for decision-making. *Advances in Neural Information Processing Systems*, *37*, pp.109460-109486.

[3] Foster, A., Ivanova, D.R., Malik, I. and Rainforth, T., 2021, July. Deep adaptive design: Amortizing sequential bayesian experimental design. In *International conference on machine learning* (pp. 3384-3395). PMLR.

[4] Huang, D., Wen, X., Bharti, A., Kaski, S. and Acerbi, L., 2026. ALINE: Joint amortization for Bayesian inference and active data acquisition. *Advances in Neural Information Processing Systems*, *38*, pp.54068-54102.

Takeaway

- *Bayesian Experimental Design (BED) provides a principled framework to select designs that can maximize the Expected Information Gain (EIG) about the parameters.*
- *Amortization can accelerate inference and design speed through pretraining.*
- *Our recent work allows Amortized BED to absorb practitioners' prior knowledge.*
- *Future work on considering healthcare foundation models as the simulator and doing decision making*

Reference

- Huang, D., Guo, Y., Acerbi, L. and Kaski, S., 2024. Amortized bayesian experimental design for decision-making. *Advances in Neural Information Processing Systems*, *37*, pp.109460-109486.
- Ward, D., Cannon, P., Beaumont, M., Fasiolo, M. and Schmon, S., 2022. Robust neural posterior estimation and statistical model criticism. *Advances in neural information processing systems*, *35*, pp.33845-33859.
- Foster, A., Ivanova, D.R., Malik, I. and Rainforth, T., 2021, July. Deep adaptive design: Amortizing sequential bayesian experimental design. In *International conference on machine learning* (pp. 3384-3395). PMLR.
- Huang, D., Wen, X., Bharti, A., Kaski, S. and Acerbi, L., 2026. ALINE: Joint amortization for Bayesian inference and active data acquisition. *Advances in Neural Information Processing Systems*, *38*, pp.54068-54102